INCH-POUND
MIL-DTL-32139/5A
2 November 2021
SUPERCEDING
MIL-DTL-32139/5
1 August 2012

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, NANOMINIATURE, SINGLE ROW, PLUG, POLARIZED SHELL, PIN CONTACTS, VERTICAL PCB THROUGH-HOLE TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32139.

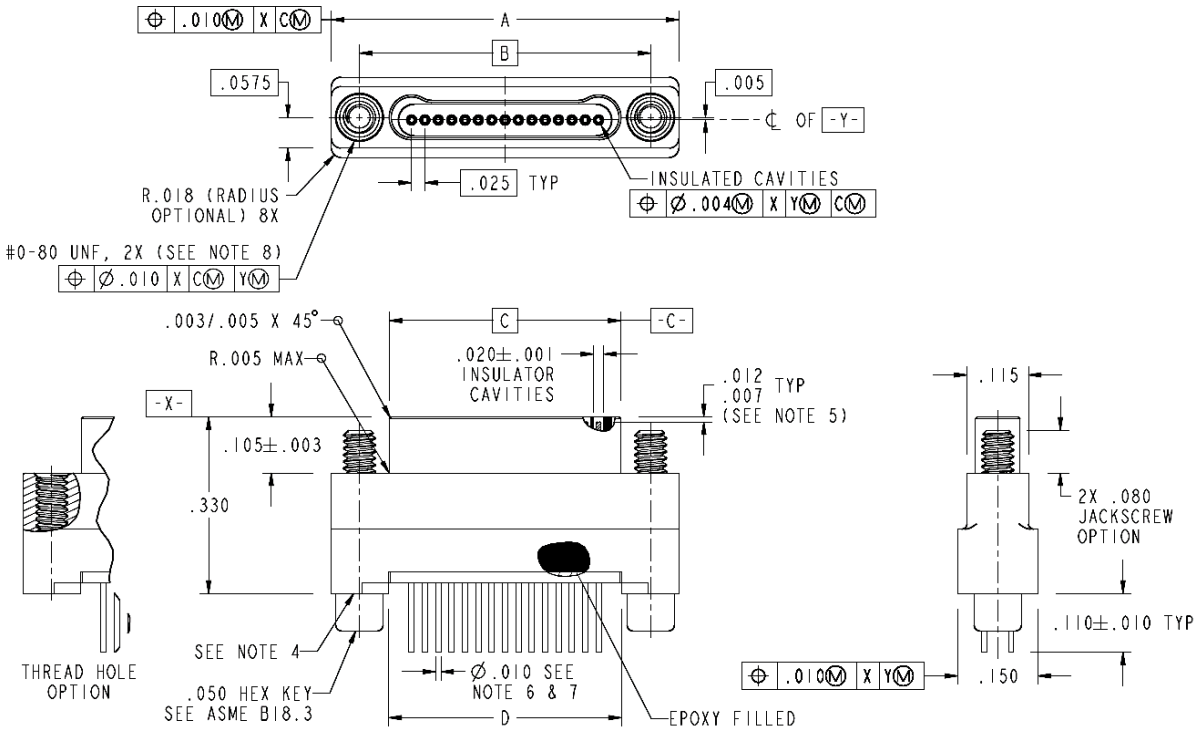


FIGURE 1. Nano connector dimensions and configurations.

AMSC N/A FSC 5935

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

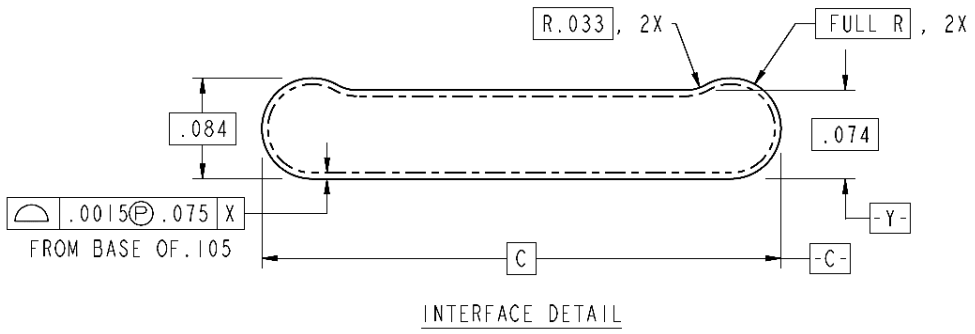
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:54Z
Check the source to verify that this is the current version before use.

Insert arrangement	A	B BSC	C BSC	D MAX
9	.500 (12.70)	.395 (10.03)	.284 (7.21)	.295 (7.49)
15	.650 (16.51)	.545 (13.84)	.434 (11.02)	.445 (11.30)
21	.800 (20.32)	.695 (17.65)	.584 (14.83)	.595 (15.11)
25	.900 (22.86)	.795 (20.19)	.684 (17.37)	.695 (17.65)
31	1.050 (26.67)	.945 (24.00)	.834 (21.18)	.845 (21.46)
37	1.200 (30.48)	1.095 (27.81)	.984 (24.99)	.995 (25.27)
51	1.550 (39.37)	1.445 (36.70)	1.334 (33.88)	1.345 (34.16)

Inches	mm
.001	0.03
.003	0.08
.004	0.10
.005	0.13
.007	0.18
.010	0.25
.012	0.30
.018	0.46
.020	0.51
.025	0.64
.050	1.27
.0575	1.46
.080	2.03
.105	2.67
.110	2.79
.115	2.92
.150	3.81
.330	8.38

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ± .005 inch (0.13 mm) angular tolerance ±2º.
4. Surface from which the lead length is measured.
5. Shell shall be flush to insulator within ± .004 inch (0.10 mm).
6. Tail finish is tin/lead or gold.
7. Leads may be flat tails that fit into the thru hole as specified in the PCB layout (Figure 4).
8. Male Threads shall be class 2A, Female threads shall be class 2B.

FIGURE 1. Nano connector dimensions and configurations – Continued.

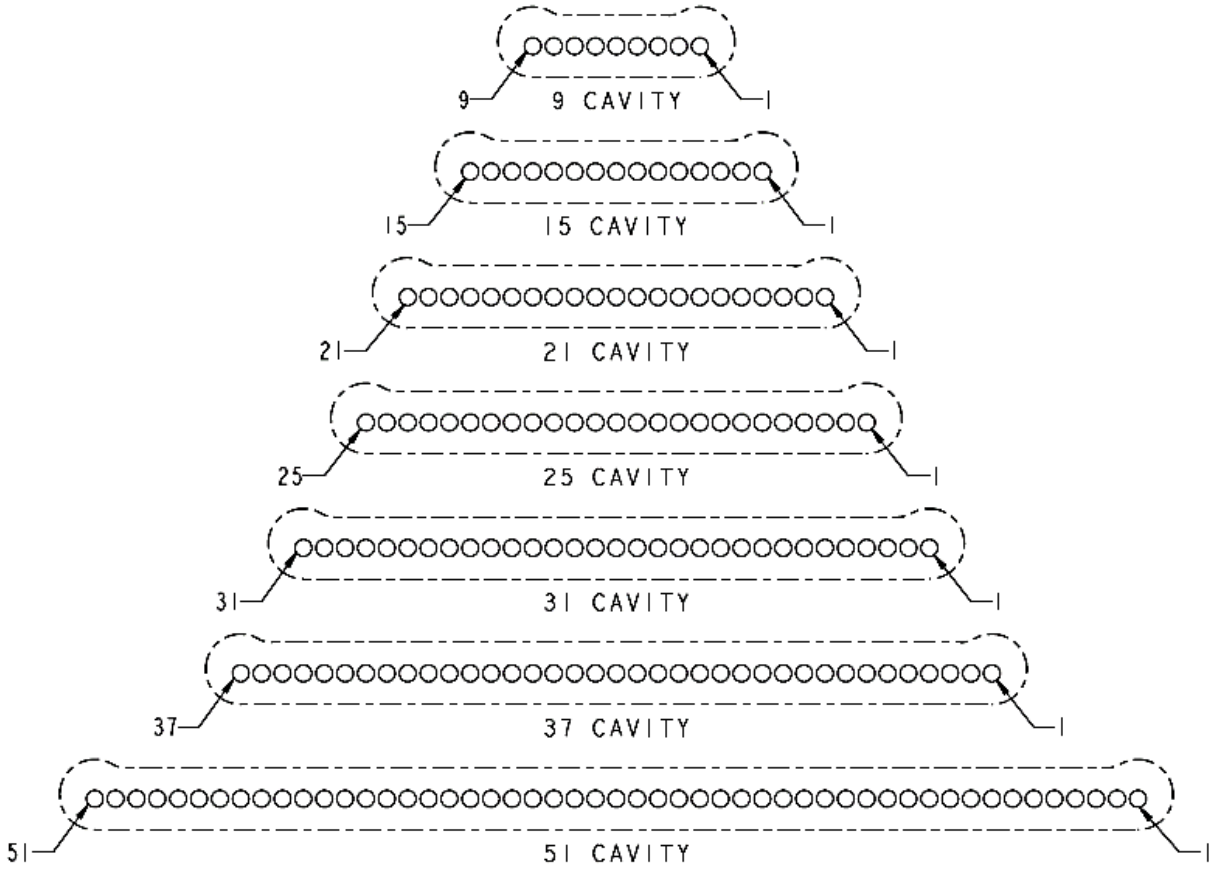


Inches	mm
.0015	0.038
.033	0.84
.074	1.88
.075	1.91
.084	2.13
.105	2.67

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ± .005 inch (.13 mm).

FIGURE 2. Nano connector interface.

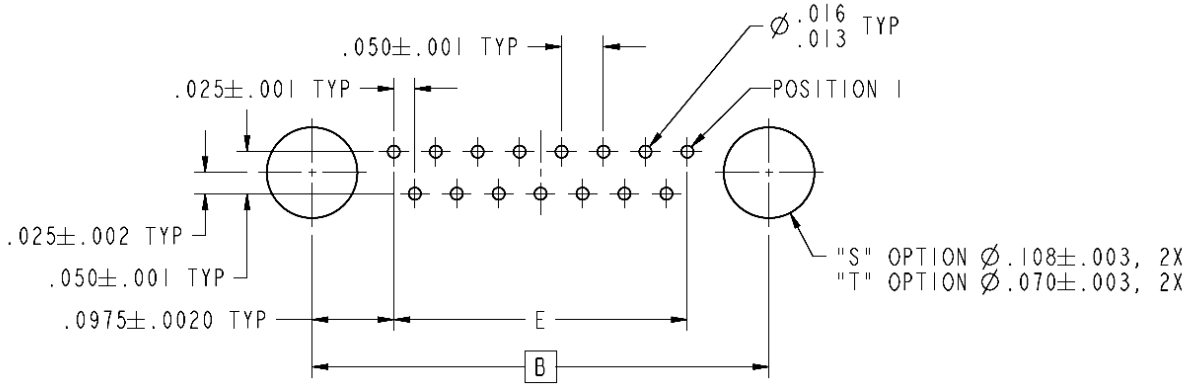
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:54Z
Check the source to verify that this is the current version before use.



NOTES:
1. Engaging face of insert shown.
2. Cavity identification numbers are for reference only and do not appear on the part.

FIGURE 3. Nano connector insert arrangement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:54Z
Check the source to verify that this is the current version before use.



Inches	mm
.001	0.03
.002	0.05
.003	0.08
.013	0.33
.016	0.41
.025	0.64
.050	1.27
.070	1.78
.0975	2.477
.108	2.74

Insert arrangement	B BSC	E
9	.395 (10.03)	.200 (5.08)
15	.545 (13.84)	.350 (8.89)
21	.695 (17.65)	.500 (12.70)
25	.795 (20.19)	.600 (15.24)
31	.945 (24.00)	.750 (19.05)
37	1.095 (27.81)	.900 (22.86)
51	1.445 (36.70)	1.250 (31.75)

NOTES:
1. Suggested hole pattern shown facing board from connector insertion side specified on figure 3.
2. Dimensions are in inches.
3. Metric equivalents are given for information only.
4. Unless otherwise specified tolerances are ± .005 inch (.13 mm) angular tolerance ±2°.

FIGURE 4. Nano connector suggested PCB hole pattern.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:54Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

Dimensions and configuration see figures 1, 2, and 3.

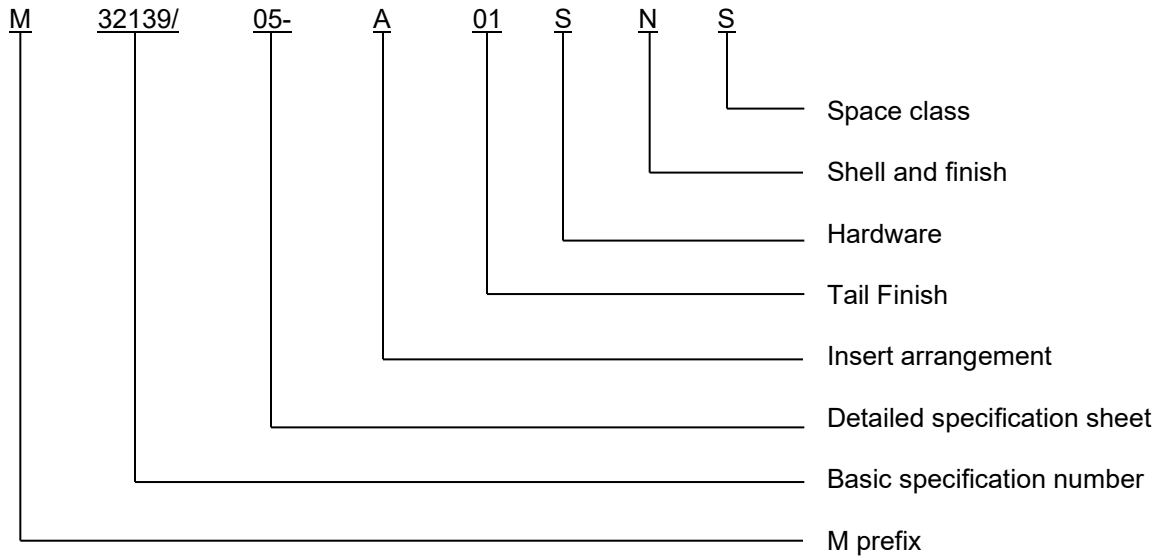
This specification sheet describes the pin side of a rectangular connector. This connector uses reverse gender contacts, i.e., the live pin is recessed in the insulator with the static socket protruding from a shrouded interface.

Contact connection: The pin contact, which is recessed in the insulator, is normally connected to the live side of the circuit.

Pins are terminated to a printed circuit board (PCB) as specified on figure 4.

Mating receptacle: Shall be in accordance with MIL-DTL-32139/2, MIL-DTL-32139/6, or MIL-DTL-32139/8.

Part or Identifying Number (PIN):



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:54Z
Check the source to verify that this is the current version before use.

Insert arrangement
A = 9
B = 15
C = 21
D = 25
E = 31
F = 37
G = 51

Tail Finish
01 = Tin/Lead
02 = Gold

Hardware [1/]
S = Jackscrew Captivated [4/]
T = Thread hole [3/]

Shell and finish
C = Aluminum shell, cadmium finish
N = Aluminum shell, electroless nickel finish [2/]
S = Passivated stainless steel shell
T = Titanium shell
A = Pure Electrodeposited Aluminum
F = Nickel Fluorocarbon Polymer
Z = Zinc Nickel

Space class
Blank for non-space applications.
S = Space class

1/ Supplied installed.
2/ When aluminum shells are required for space applications, electroless nickel shall be used. Cadmium finish is not acceptable (see MIL-DTL-32139).
3/ The aluminum shell may require additional thread component which may affect the minimum and maximum mounting screw length. Check with manufacture for recommended mounting screw length.
4/ The only mechanical retention for connectors with jackscrew option, is the collective solder joints of the thru hole leads. Consider designing other means of fastening, i.e. epoxy.

Alternate shell finishes:

Pure Electrodeposited Aluminum. Pure dense electrodeposited aluminum shall be in accordance with MIL-DTL-83488, type II shall withstand 48 hour salt spray. Color shall be non-reflective and shall meet the requirements as specified herein.

Nickel Fluorocarbon Polymer. High phosphate nickel with fluorocarbon polymer additive over a suitable underplate shall be able shall withstand 48 hour salt spray. Color shall be non-reflective and shall meet the requirements as specified herein.

Zinc Nickel. Zinc Nickel Alloy shall be in accordance with ASTM B841, over a suitable underplate shall withstand 48 hour salt spray. Color shall be non-reflective and shall meet the requirements as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:54Z
Check the source to verify that this is the current version before use.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-32139, this document references the following:

MIL-DTL-32139/2
MIL-DTL-32139/6
MIL-DTL-32139/8
MIL-DTL-83488
ASME B18.3
ASTM B841

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AV, MI

Preparing activity:
DLA - CC

(Project 5935-2021-126)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:54Z
Check the source to verify that this is the current version before use.